

Soundharya Shanmugavel

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Professional Summary

Innovative AI Developer with 4+ years of overall experience, including 2+ years specializing in GenAI and AI-driven automation. Expert in designing multi-agent systems using frameworks like CrewAI, AutoGen, LangGraph, Azure, and AWS Bedrock. Proven ability to transform manual workflows into intelligent and scalable systems that enhance operational efficiency and user experience. Extensive experience in Infrastructure as Code (Terraform, CloudFormation), RAG implementations, and low-code platforms such as Appian, OutSystems, and Microsoft Power Platform. Recognized multiple times as "Spotlight of the Team" for technical excellence and impactful project execution.

Work Experience

Technical Consultant

October 2021 – Present

WNS-Vuram

- Promoted twice within 4 years from Intern → Associate Technical Consultant → Technical Consultant based on technical excellence and consistent delivery
- Awarded "Spotlight of the Team" multiple times for technical excellence and innovative AI implementations
- Received special recognition award for innovative AI implementation in democratizing whitepaper generation
- Designed & developed AI-powered applications using CrewAI, n8n, AutoGen, LangGraph, and AWS Bedrock for automation and decision-making
- Built LLM-based AI agents for workflow automation, enhancing efficiency across multiple business processes
- Integrated AI-driven solutions with cloud services (Azure AI, AWS Lambda, Firebase Firestore) for scalable deployments
- Automated repetitive tasks using Python scripting & no-code/low-code tools (Appian, OutSystems, PowerApps), reducing manual effort
- Worked with multi-agent collaboration frameworks to create interactive and adaptive AI applications
- Implemented Infrastructure as Code using Terraform and CloudFormation for scalable cloud deployments
- Containerized AI applications and deployed to Azure Container instances using Docker
- Developed RAG (Retrieval-Augmented Generation) systems for enhanced AI knowledge retrieval

Key Projects

AgentOpsAI - AI Agent Observability & Governance Platform

Bengaluru, Karnataka

WNS-Vuram

August 2025 – Present

- Architected a platform for real-time observability, governance, and quality assurance across the full AI agent lifecycle, from registration and testing to deployment and operations.
- Implemented automated prompt optimization, responsible AI checks, guardrail recommendation and evaluation analytics to continuously monitor agent performance, correctness, and compliance.
- Developed features for cost analysis, A/B testing, and real-time alerting on latency, quality degradation, token anomalies, and system failures.
- Built natural language business dashboard that interprets user queries and searches databases to provide insights
- Utilized a tech stack including Python, AWS Bedrock, Lambda, DynamoDB, S3, EventBridge, Claude and React.

VybeStudio - AI-Powered Product Development Accelerator

Bengaluru, Karnataka

WNS-Vuram

August 2025 – September 2025

- Architected an innovative AI-powered solution to accelerate product development by automating requirement gathering, documentation, and agile workflows

- Developed intelligent agents that generate comprehensive Business Requirements Documents (BRDs), epics, user stories, and Standard Operating Procedures (SOPs) from existing codebase analysis
- Implemented automated workflow generation that transforms technical specifications into actionable development tasks
- Deployed the solution using Azure Container instances with Docker for scalable and reliable performance
- Reduced product development documentation time by 70% while ensuring consistency and quality across all generated artifacts

LLM as a Judge - AI-Powered Evaluation System

*Bengaluru, Karnataka
July 2025 – August 2025*

WNS-Vuram

- Developed a sophisticated AI-powered evaluation system to analyze and assess the performance of agent and multi-agent systems
- Implemented intelligent scoring mechanisms that evaluate agent responses, decision-making processes, and overall system effectiveness
- Designed automated benchmarking protocols to compare different AI models and configurations
- Deployed the system using Azure Functions for serverless, cost-effective operation with automatic scaling
- Created comprehensive evaluation metrics and reporting dashboards for continuous system improvement

CDMS (Card Dispute Management System) - AI-Driven Dispute Resolution

*Bengaluru, Karnataka
June 2025 – July 2025*

WNS-Vuram

- Engineered an AI-driven dispute management system to automate the end-to-end lifecycle of card dispute handling for financial institutions
- Developed intelligent workflow automation that processes dispute claims, analyzes evidence, and generates resolution recommendations
- Implemented smart categorization and priority scoring for disputes based on complexity, amount, and historical patterns
- Created automated communication systems for status updates and resolution notifications to customers and stakeholders
- Reduced dispute resolution time by 60% while maintaining regulatory compliance and improving customer satisfaction

IDP Agent - Intelligent Document Processing System

*Bengaluru, Karnataka
May 2025 – June 2025*

WNS-Vuram

- Developed an advanced Intelligent Document Processing (IDP) agent capable of classifying and extracting specific fields from various document types
- Implemented machine learning models for automatic document type identification and intelligent field extraction
- Created adaptive processing pipelines that handle structured, semi-structured, and unstructured documents with high accuracy
- Designed validation mechanisms to ensure data quality and accuracy in extracted information
- Achieved 95% accuracy in document classification and field extraction across multiple document formats

TBML (Trade-Based Money Laundering) Detection Platform

2023 – 2024

- Worked as a PowerApps developer in the creation of an RPA automation for Trade Based Money Laundering manual processes for a banking client
- Designed the automation using Power Automate RPA, Power Automate Cloud flows, and a PowerApps canvas application
- Implemented features to track automation executions within the PowerApps canvas application
- Reduced manual compliance checking effort by over 60% while enhancing traceability
- Implemented secure data handling protocols meeting strict financial regulatory requirements

HireX AI - Advanced Hiring Automation System

*Bengaluru, Karnataka
October 2024 – November*

WNS-Vuram

2024

- Architected an advanced hiring automation system leveraging 5 specialized AI agents working in coordination
- Developed intelligent document processing agents that scan multiple resumes simultaneously, extracting key information such as skills, experience levels, and other critical candidate details
- Implemented a comprehensive workflow where AI agents analyze candidates against job descriptions, store structured data in databases, identify optimal matches, and automatically distribute personalized selection emails
- Reduced hiring cycle time by 65% while improving candidate-job matching accuracy through AI-driven evaluation

Dashboard Accelerator AI - Multi-Agent Dashboard Generation System

WNS-Vuram

Bengaluru, Karnataka
October 2024 – November
2024

- Engineered a sophisticated system utilizing 4 specialized AI agents that collaborate to transform raw data into interactive HTML, CSS, and JavaScript dashboards
- Integrated Langchain's CODE INTERPRETER as a custom tool, empowering AI agents to analyze data patterns and generate optimized visualizations
- Designed an intelligent workflow that processes Excel/CSV files by analyzing data context and dashboard requirements to automatically produce tailored KPIs, standard charts, and complex data visualizations
- Reduced dashboard development time by 80% compared to traditional methods while maintaining professional quality standards

AWS Bedrock Multi-Agent System

WNS-Vuram

Bengaluru, Karnataka
February 2025

- Architected and developed a scalable multi-agent system using AWS Bedrock foundation models
- Deployed serverless architecture utilizing AWS Lambda functions for cost-effective and responsive AI processing
- Implemented agent coordination patterns enabling complex task completion through specialized agent interactions
- Designed and documented a reusable framework for rapidly developing new agent-based applications on AWS infrastructure

Whitepaper Democratization System

2023 – 2024

- Created a multi-agent system that generates whitepaper integrated with frontend
- Engineered an innovative multi-agent system leveraging LangGraph, AutoGen, and LangChain
- Designed prompt engineering protocols to ensure high-quality content generation
- Created seamless integration between AI backend and React-based frontend interface
- Reduced whitepaper creation time by 80% while maintaining quality standards

Technical Expertise

GenAI & AI Development: CrewAI, n8n, AutoGen, LangGraph, LangChain, AWS Bedrock, LLM-based AI development, AI Agents, Workflow Automation, OpenAI API, Azure AI, Firebase ML, Prompt Engineering, RAG (Retrieval-Augmented Generation)

Software Development: Python, JavaScript, React, SQL, Docker, Containerization, CI/CD, AWS Lambda, Azure Container Instances, Azure Functions

Infrastructure & DevOps: Terraform, CloudFormation, Infrastructure as Code (IaC), Docker, Azure Container Instances, AWS Lambda, Serverless Architecture

Low-Code/No-Code: Microsoft PowerApps, Power Automate, Appian, OutSystems, Microsoft 365, Agile Methodologies

Cloud Services: Azure services, AWS services, Firebase Firestore, Azure AI, AWS Bedrock

Certifications

- Microsoft Certified: Power Platform Developer Associate
- Microsoft Certified: Power Platform App Maker Associate
- Microsoft Certified: Azure AI Fundamentals

Education

Bachelor of Engineering, Computer Science

National Engineering College

August 2018 – May 2022

- CGPA: 8.7/10

Languages

Tamil: Native or Bilingual Proficiency

English: Professional Working Proficiency